

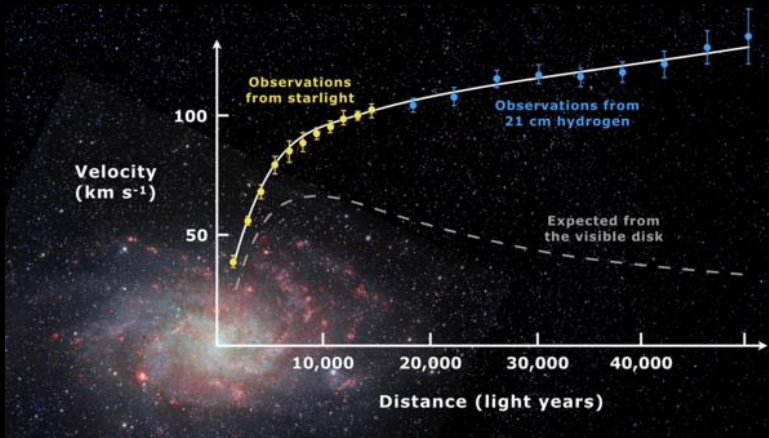


Hunting For Dark Matter with LDMX

Kieran Wall



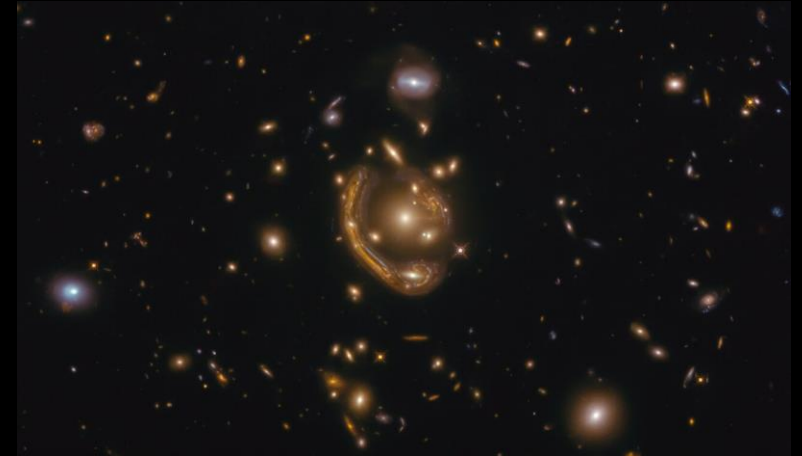
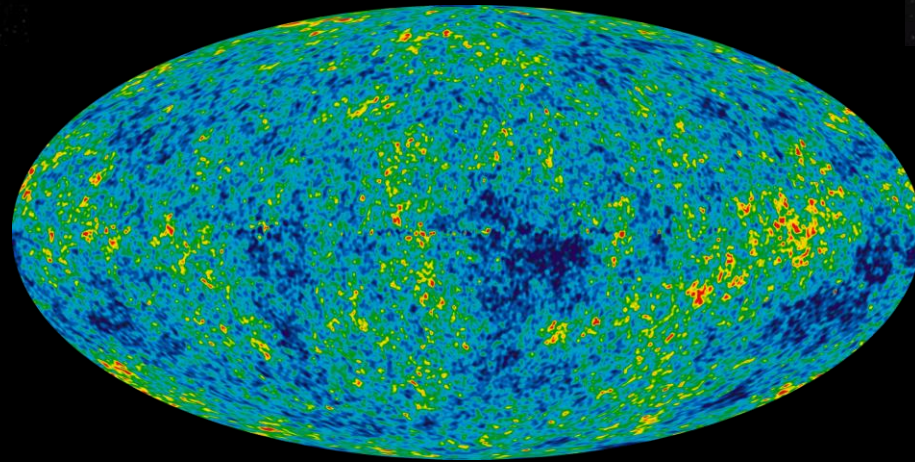
Dark Matter is out there!



Galaxy Rotation Curves

Can look for particle
dark matter here on
Earth!

Cosmic Microwave Background



Gravitational Lensing

Thermal Dark Matter

Mass



- Candidate masses span ~70 orders of magnitude!

Simplifying assumption: Dark matter was in thermal equilibrium in the early universe

- Well motivated
- Dramatically reduced parameter space



- Lots of open parameter space ☺
- The focus of LDMX
- Restricted parameter space ☹

LDMX Concept

- Light DM could theoretically interact with SM matter via a “dark photon” (A')
- Potential interaction: Dark Bremsstrahlung
 - Electron slows down around atom and emits an A' instead of a photon!
- A' behavior dependent on mass relative to the light dark matter particle $\rightarrow$

If $m_{A'} > 2m_{\text{DM}}$ then A' decays *invisibly*

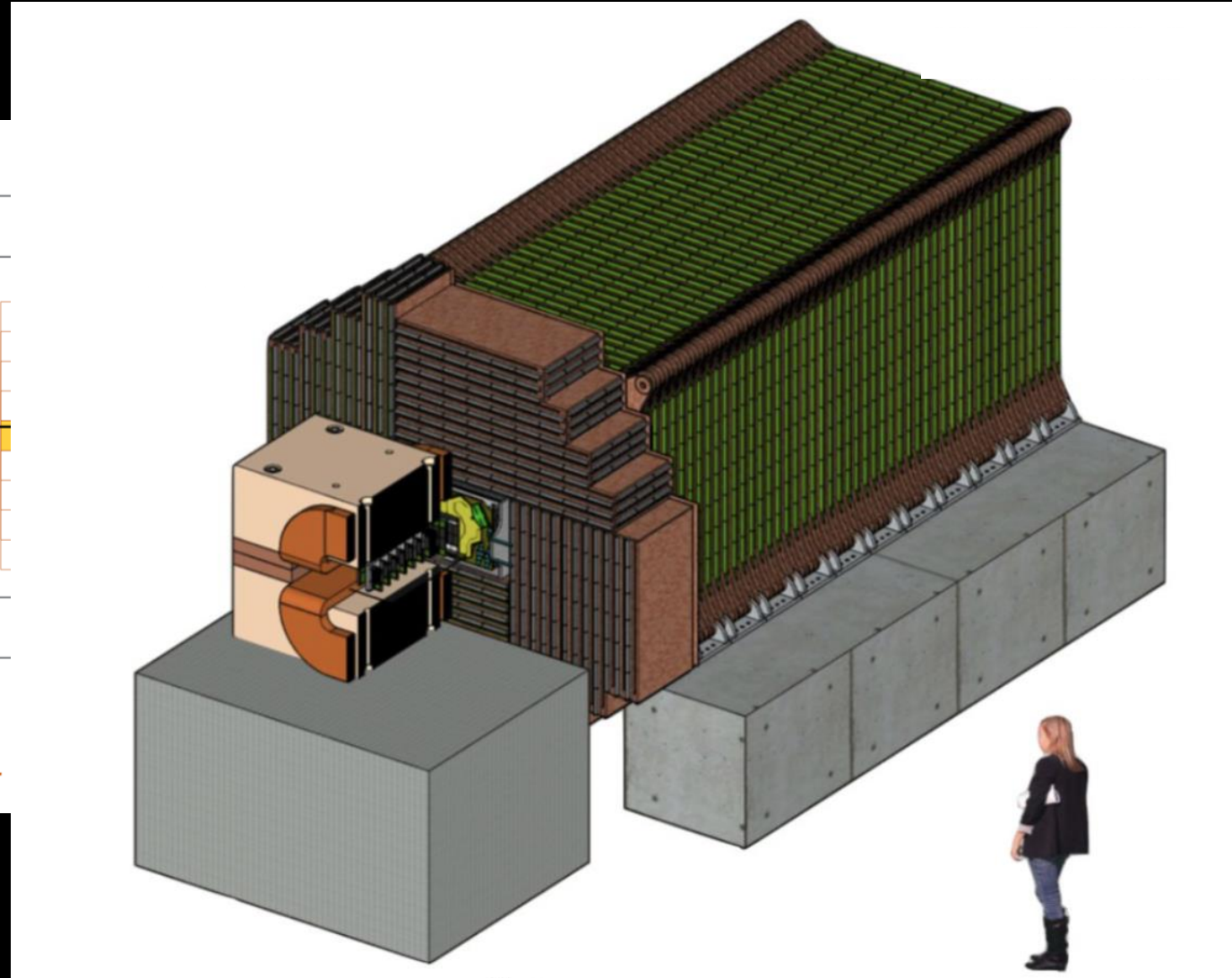
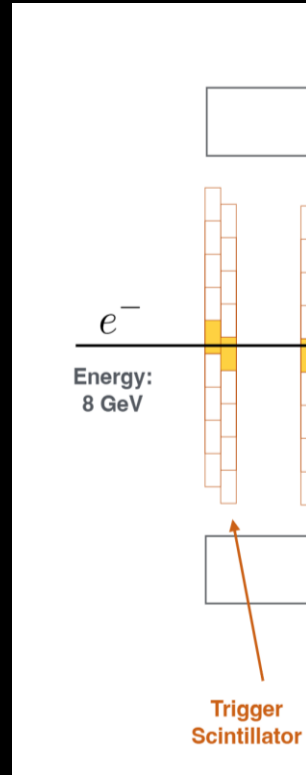
“Missing Momentum”
Search $\rightarrow$ Primary LDMX
Search

If $m_{A'} < 2m_{\text{DM}}$ then A' decays *visibly*

My Primary Research
Interest!
(Focus of rest of talk)

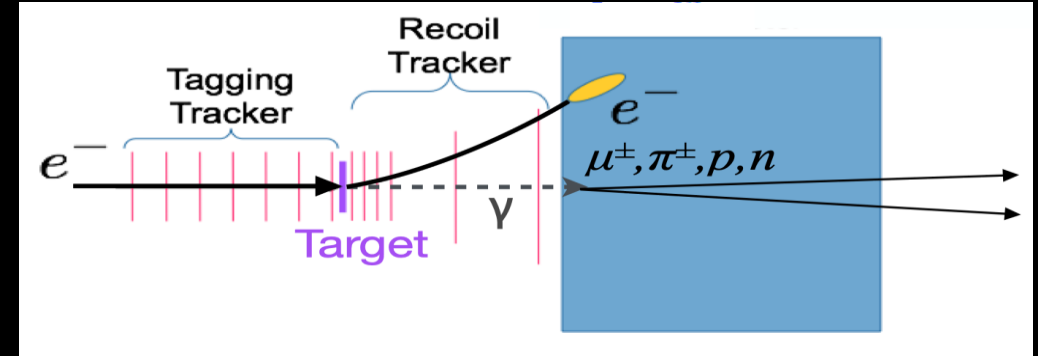
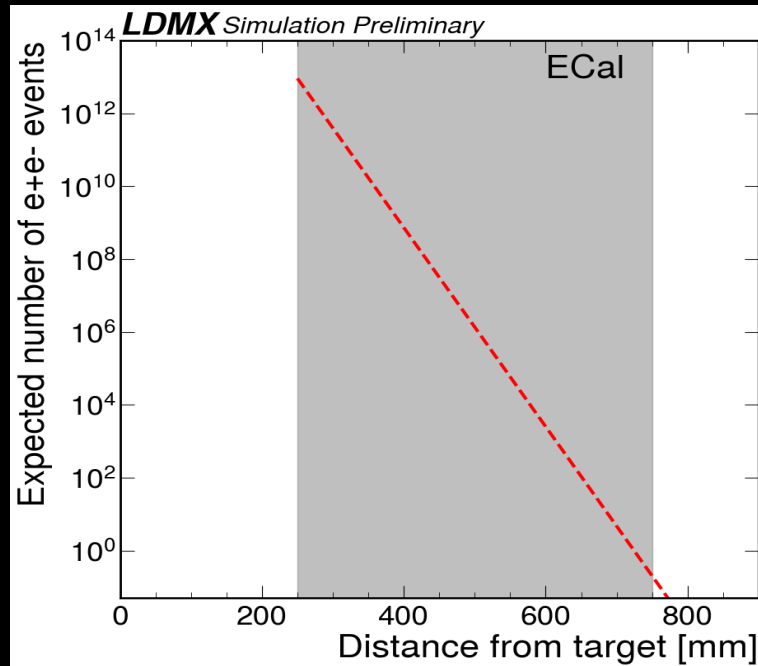
LDMX Design

- High-rate electron beam (from SLAC) at Tungsten target to maximize potential dark-brem events
- Series of detectors and calorimeters can carefully look for displaced electromagnetic showers



LDMX Visibles -> Backgrounds

- Could in theory look for visible A' in Ecal and Hcal:
 - Simulations revealed irreducible $\gamma \rightarrow e^+e^-$ background
 - Restrict search to Hcal



- Hcal backgrounds:
 - Rare photonuclear events
 - $\gamma \rightarrow \mu^+\mu^-$ events

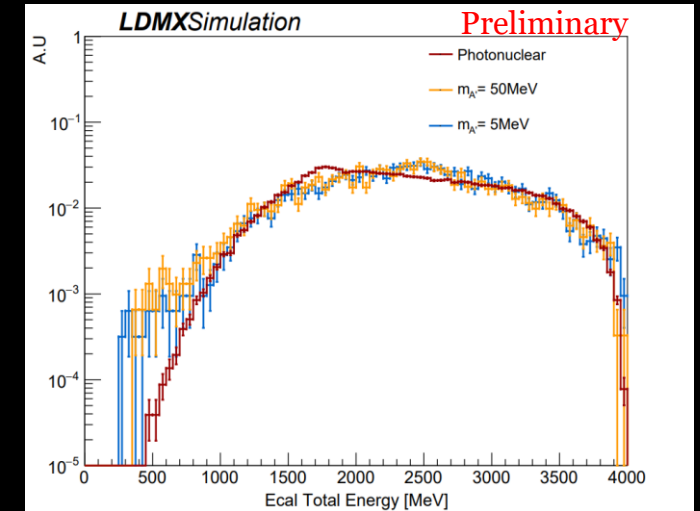
Carefully study
differences in events
features between signal
and background!

Visibles BDT

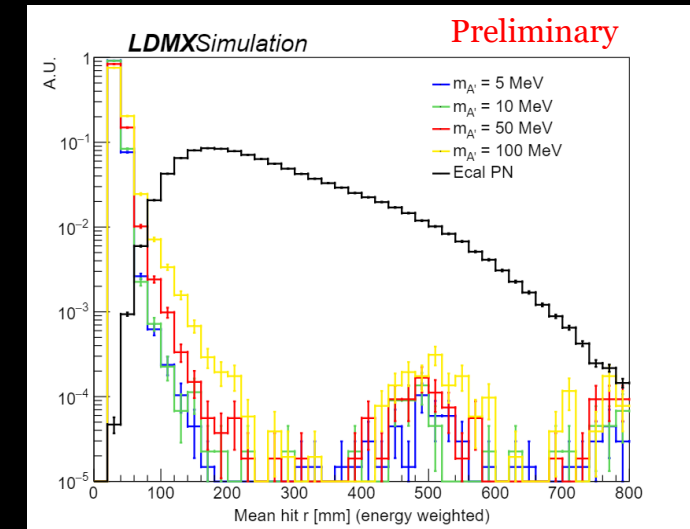
- Some event features have clear difference between signal and background
- In others signal and background very similar
- Could make threshold cuts on single features with good discrimination...

Or even better
combine multiple
high performing
features with BDT!

BDT = Boosted Decision Tree -> Simple but powerful ML model

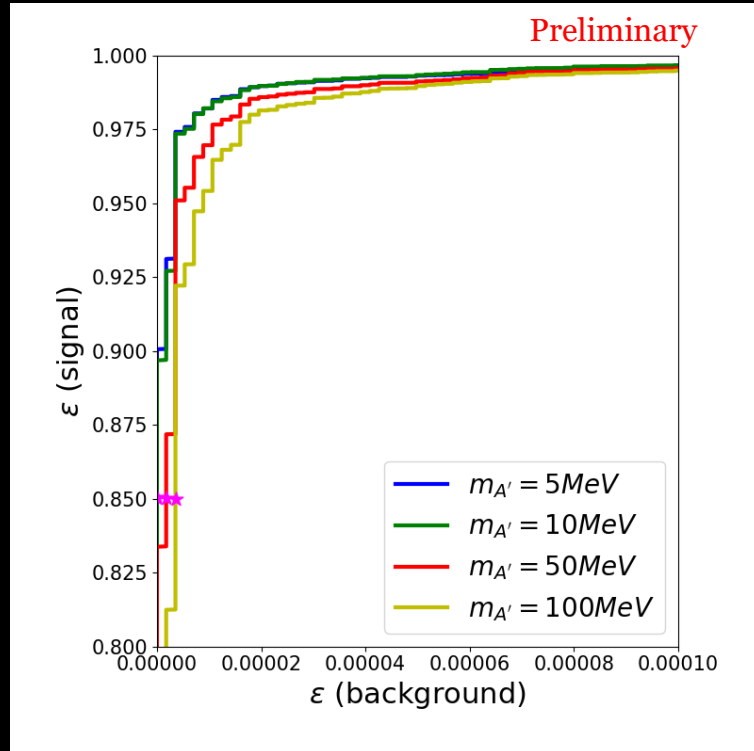


Bad Input Feature ☹️



Good Input Feature 😊

BDT Performance

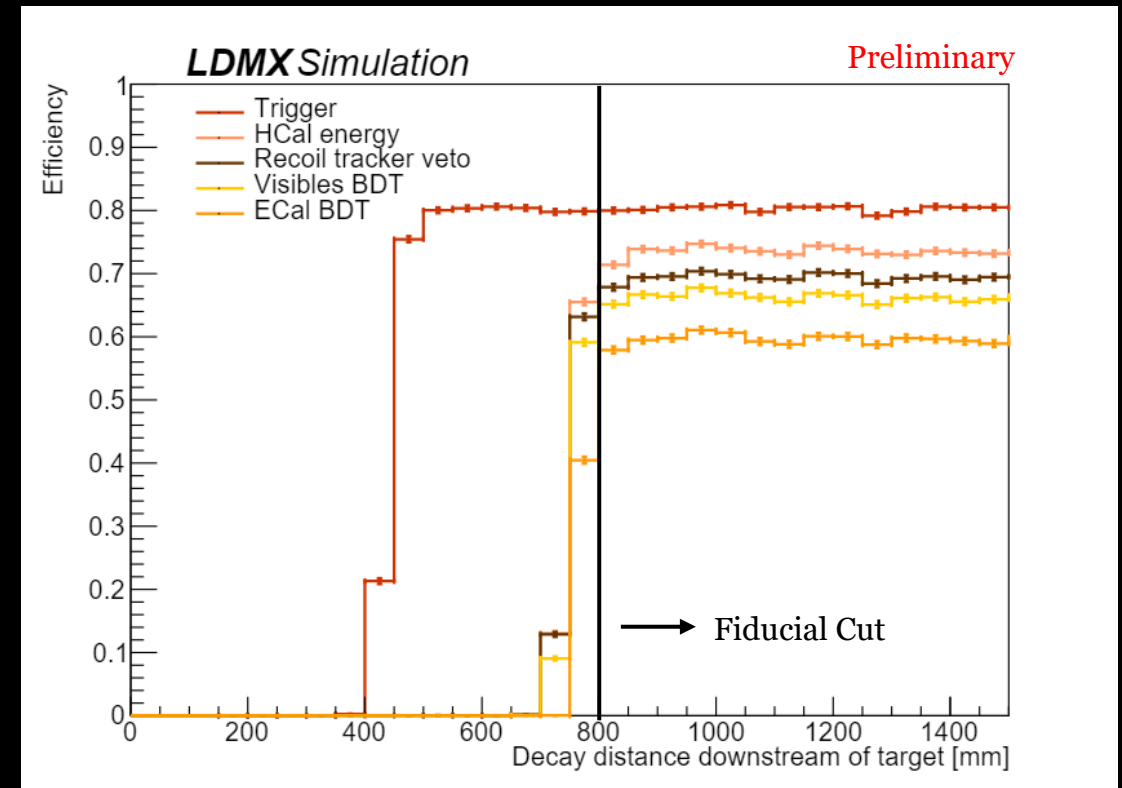


Have some more veto handles to get to zero background!

- BDT combines 12 Hcal discriminating features, trained on simulated signal and background events
- Tested against 10^{14} EoT equivalent background sample
 - Preliminary cuts +
Visibles BDT = **189 passing events**
 - Strong signal efficiency

Further Veto Handles

- Missing Momentum search uses a BDT trained on Ecal features, allows us to incorporate information about recoil e^- shower:
 - Preliminaries + Visibles BDT + Ecal BDT = **5 passing events**
- If we require shower in the Hcal starts beyond first layer, we can reach **zero background events** at 10^{14} EoT. Cool!
 - We maintain high signal efficiency throughout whole cutflow



Conclusion

- LDMX Visibles offers complementary search channel for LDMX with potential for zero background and competitive signal efficiency!



Thank You!

Questions?



State-of-the-art LDMX prototype



Our International Collaboration

My research this summer was funded by UVA's Mitchell Fellowship – Thanks!

Image Sources

- https://en.wikipedia.org/wiki/File:Standard_Model_of_Elementary_Particles.svg
- [https://en.wikipedia.org/wiki/Galaxy_rotation_curve#/media/File:Rotation_curve_of_spiral_galaxy_Messier_33_\(Triangulum\).png](https://en.wikipedia.org/wiki/Galaxy_rotation_curve#/media/File:Rotation_curve_of_spiral_galaxy_Messier_33_(Triangulum).png)
- <https://esahubble.org/wordbank/gravitational-lensing/>
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